



యోగి వేమన విశ్వవిద్యాలయం, కడప
YOGI VEMANA UNIVERSITY



Re-accredited by NAAC 'A' Grade with CGPA of 3.13/4.0 (2nd Cycle)

Department of Zoology

**Two Year M.Sc. Courses (4 SEMESTERS) with effect
from the academic year 2025-26 as per NEP -2020**

Dr. S. P. Venkata Ramana
Head & BOS Chairman
Department of Zoology
Yogi Vemana University
Kadapa - 516005



YOGI VEMANA UNIVERSITY

DEPARTMENT OF ZOOLOGY

Vemanapuram, Kadapa-516005, Andhra Pradesh, India.

17/07/2025

Minutes of the meeting of Board of studies council held on 17/07/2025 at 3:00 P.M. VIRTUAL MODE in ROOM NO: 325 (Second M.Sc., class room), Department of Zoology, Yogi Vemana University, Kadapa.

Members present

1. Dr. S.P. Venkata Ramana
2. Prof. M. Rajasekhar
3. Prof. A. Bindu Madhava Reddy
4. Dr. N.V.Rami Reddy
5. Dr. V. Anu Prasanna
6. Dr. A.T.Venkatramana Reddy
7. Ms.P. Hemavardhini

Designation

- Chairman, BOS
External Member
External Member
Member
Member
Member
Student Member

Agenda:

1. Restructuring of two years M.Sc., Zoology (4 Semesters) and MSc., Lateral entry (2 Semesters) course patterns, syllabus, scheme of examination and modalities for implementation with effect from the academic year 2025 – 2026 as per NEP 2020 norms.

Resolution:

It is resolved to restructure the two years MSc., Zoology (4 Semesters) and MSc., Lateral entry(2 Semesters) course patterns, syllabus, scheme of examination and modalities for implementation with effect from the academic year 2025 – 2026 approved as per NEP 2020 norms with the university guidelines.

Members present

Designation

Signature

1. Dr. S.P. Venkata Ramana Chairman, BOS
2. Prof. M. Rajasekhar External Member
3. Prof. A. Bindu Madhava Reddy External Member
4. Dr. N.V.Rami Reddy Member
5. Dr. V. Anu Prasanna Member
6. Dr. A.T.Venkatramana Reddy Member
7. Ms.P. Hemavardhini Student Member

S.P.V. Venkata Ramana
17/7/2025

M. Rajasekhar

A. Bindu Madhava Reddy

N.V. Rami Reddy
17/7/25

V. Anu Prasanna
17/7/25

A.T. Venkatramana Reddy

Ms.P. Hemavardhini



REGULATIONS

Two Year **M.Sc. Courses (4 SEMESTERS)** with effect from the academic year 2025-26 as per
NEP -2020

Preamble

The National Education Policy (NEP) 2020 recognizes that higher education plays an extremely important role in promoting human and societal well-being and in developing India as envisioned in its constitution—a democratic, just, socially conscious, cultured, and humane nation up holding liberty, equality, fraternity, and justice for all. It notes that “given the 21st-century requirements, quality higher education must aim to develop good, thoughtful, well-rounded, and creative individuals”.

As per NEP 2020, the UGC has formulated a new student-centric “Curriculum and Credit Framework for Undergraduate Programmes (CCFUP)” incorporating a flexible choice-based credit system, a multidisciplinary approach, and multiple entry and exit options. This will facilitate students' pursuit of their career path by choosing the subject/field of their interest.

In consonance with NEP 2020, the Government of Andhra Pradesh rolled out a redesigned CBCS curriculum for 4-year UG Honors Programmes from the year 2020-21. With the CCFUP guidelines released in December 2022, the AP State Council of Higher Education redesigned the curricular framework for the UG Degree Programmes, by introducing a 4-Year UG Program with single major and minors with multiple entries and multiple exits from the academic year 2023-24. The Yogi Vemana University, Kadapa, reframed the guidelines and course structure per the UGC, New Delhi, and APSCHE guidelines from 2025-26.

1. Eligibility

A candidate seeking admission as per National Education Policy-2020 into the following 2-year (4 Semesters) Post-graduate courses in table below should have passed the qualifying bachelor degree examination of any UGC recognized university as equivalent there to, as mentioned against that course in table below. Besides this, the candidate has to qualify in the prescribed entrance examination of that year conducted by YVU/ APSCHE/any Apex body. Spot admission will also be carried out with the permission from the apex body.

The eligibility criteria as follows

1.1. A B.Sc. degree with Honors/Honors with Research for 1 year/2-semesters M.Sc. (Lateral



Entry).

- 1.2. A three-year 6-semester B.Sc. degree for a 2-year/4-semester M.Sc.
- 1.3. A student must have studied either a major or a minor discipline in the UG program

Code	Name of the Course	Qualifying degree for admission
01	Applied Mathematics	Bachelor's Degree with Mathematics as major/minor discipline
02	Biochemistry	Bachelor's Degree with Biochemistry/ any Biology subject as major/minor discipline.
03	Biotechnology	Bachelor's Degree with Biotechnology/ any Bio Science & Technology subjects as major/minor discipline.
04	Botany	Bachelor's Degree with Botany/ any Biology subject as major/minor discipline.
05	Chemistry	Bachelor's Degree with Chemistry/ Chemical Sciences as major/minor discipline.
06	Computer Science	Bachelor's Degree with Computer Science as major/minor discipline.
07	Environmental Science	Bachelor's Degree with any Science as major/minor discipline.
08	Genetics and Genomics	Bachelor's Degree with any Biology as major/minor discipline.
09	Geology	Bachelor's Degree with Geology/ any Science & Technology subjects as major/minor discipline.
10	Material Science & Nanotechnology	Bachelor's Degree with Chemistry/Physics subjects as major/minor discipline.
11	Microbiology	Bachelor's Degree with Microbiology/ any Biology subject as major/minor discipline.
12	Physics	Bachelor's Degree with Physics subject as major/minor discipline.
13	Psychology	Any Bachelor's Degree
14	Food Science and Technology	Bachelor's Degree with Food Science/ Food Technology/Home Science/ Nutrition/ any biology subject as major/minor discipline.
15	Zoology	Bachelor's Degree with Zoology/ any Biology subject as major/minor discipline.

1.4. Sponsored Seats

2. Lateral Entry and Exit Options

2.1. Lateral Entry

A student who has completed a 4-year UG Honors Degree and a 4-year UG Honors with



Research is eligible for a lateral entry into the 2nd year of the program, with course work in the 3rd semester and project work in the 4th semester.

2.2. Exit Point

For the PG programme, there shall only be one exit point for those who join two-year PG program. The exit option for a two-year M.Sc. may be offered at the end of the first year after successful completion of the two semesters and the student may be awarded a PG Diploma on request.

2.3. Supernumerary Seats for UG Honors candidates

The total number of seats sanctioned for the M.Sc., programme shall remain the same, without increasing the sanctioned strength. However, for the admission of students with 4-year UG Honours and 4-year UG Honours with Research, 20% of supernumerary seats shall be sanctioned in the 3rd semester of the 2nd year. Such supernumerary seats created shall be used only for the purpose of sanctioned without any deviation. No supernumerary seats shall be made if no suitable candidates are found. The unfilled seats of the sanctioned strength of the program can be filled with the lateral entry candidates in the 2nd year with the prior permission from the University.

3. Course of Study

3.1. For those who joined with a 3-year UG programme, the M.Sc. is 2 Years with 4 semesters, each with 90 working days of six hours daily.

3.2. The program duration for those joining with a 4-year UG Honours/Honours with Research Degree shall be one year with two semesters, each lasting 90 working days at six hours a day.

3.3. The working days include the days allotted for tests, examinations, and preparation holidays, and the inter-semester break shall be one week.

3.4. The Course of study for the M.Sc., degree shall be according to the curriculum frame, types of courses to be offered, course structure, syllabus, list of text, and reference books prescribed by the University from time to time.

4. Curricular Framework

The NEP 2020 recommended National Higher Education Qualification Framework (NHEQF) shall be followed for the course framework. The framework shall also be in sync with the National Credit Framework (NCrF) for the creditization of assessment. A three-year, B.Sc., (six-semester's) student can pursue a two-year M.Sc., (four-semester's) with course work in the first three semesters and project work in the fourth semester of the second year, which is equivalent to



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84 credits (57 credits for theory and 27 credits for practical's). Further, B.Sc., with Honors or Honors with Research student can pursue a one-year, two-semester M.Sc., with course work in the third semester and project work in the fourth semester, which is equivalent to 40 credits (20 credits for theory and 20 credits for practical's). The details are as mentioned below;



DEPARTMENT OF ZOOLOGY

MSc, ZOOLOGY SYLLABUS (NEP -2020) CURRICULAR FRAME WORK

(WEF 2025-2026)

1.A. Redesigned curricular frame work of 2-year post graduate program with course work in first 3 semesters and project work in 4th semester of 2 year

2-Year Master of Science (M.Sc.) with course work in first 3 semesters and project work in 4 th semester of II year							
Semester	Courses & Course code	Level of Course	Title of the Course	Theory		Practical	
				H	Cr	H	Cr
Sem1	Core Course 1: ZOO CC-101	500 - 599	Structural Anatomy of Invertebrata & Vertebrata	4	3	3	1
	Core Course 2: ZOO CC-102	500 - 599	Genetics & Evolution	4	3	3	1
	Core Course 3: ZOO CC-103	500 - 599	Metabolic Regulation & Cell Function	4	3	3	1
	Skill Oriented Course 1: ZOO SOC-104	400 - 499	Tools and Techniques in Biology	3	3	3	1
	Skill Oriented Course 2: ZOO SOC-105	400 - 499	Field research techniques for wildlife conservation	3	3	3	1
	Open Online Transdisciplinary Course 1 (Students can choose a 4-week Course from the courses available in the SWAYAM & SWAYAM PLUS) https://swayam.gov.in/ https://swayam-plus.swayam2.ac.in/ / Offline Open Elective Course offered by University ZOO-OOTC/OE-106	300 - 399			2		
	Indian Knowledge Systems 1	200 - 299	Indian Knowledge Systems (IKS) –An	2	0	AUDIT COURSE	



	(Either the department can design a 2-credit course or may choose from online https://iksindia.org/courses-offered-by-iks-centers.php): ZOO-IKS-107		overview				
	End of Sem1			20	17	15	5
	Core Course 4: ZOO CC-201	500 - 599	Molecular Cell Biology	4	3	3	1
	Core Course 5: ZOO CC-202	500 - 599	Developmental Biology	4	3	3	1
	Core Course 6: ZOO CC-203	500 - 599	Comparative Animal Physiology	4	3	3	1
	Skill Oriented Course 3: ZOO SOC-204	400 - 499	Toxicology & Immunobiology	3	3	3	1
	Skill Oriented Course 4: ZOO SOC-205	400 - 499	Biostatistics & Bioinformatics	3	3	3	1
Sem2	Open Online Transdisciplinary Course 2 (Students can choose a 4 weeks Course from the courses available in the ONLINE SWAYAM & SWAYAM PLUS) https://swayam.gov.in/ https://swayam-plus.swayam2.ac.in/ : ZOO-OOTC-206	300 - 399			2		
	Indian Knowledge Systems 2 (Either the department can design a 2-credit course or may choose from online https://iksindia.org/courses-offered-by-iks-centers.php): ZOO-IKS-207	200 - 299	Contribution of IKS to Science and Technology	2	0		AUDIT COURSE
	End of Sem2			20	17	15	5
	Core Course 7: ZOO CC-301	500 - 599	Neurobiology	4	3	3	1
	Core Course 8: ZOO CC-302	500 - 599	Applied Animal	4	3	3	1



Sem3			Biotechnology				
	Core Course 9: ZOO CC-303	500 - 599	Ecology and Economic Zoology	4	3	3	1
	Skill Oriented Course 5: ZOO SOC-304	400 - 499	Aquaculture and Health Management	3	3	3	1
	Skill Oriented Course 6: ZOO SOC-305	400 - 499	Endocrinology & Enzymology	3	3	3	1
	End of Sem3			18	15	15	5
Sem4	Open Online Skill Development Course/s (Students can choose an 8-credit course or 4 Course from the courses available in the ONLINE SWAYAM & SWAYAM PLUS) https://swayam.gov.in/ https://swayam-plus.swayam2.ac.in/ https://onlinecourses.swayam2.ac.in/ntr25_ed27/preview w: ZOO-SDC-401	300 - 399		-	8	-	-
	Project Work: ZOO-PW-402	400 - 499					12
	End of Sem4			-	8	-	12
Total Hours/Credits				58	57	45	27



5. Types of Courses to be Offered

5.1. Core Courses

The PG program with course work in the first 3 semesters and project work in 4th semester of 2nd year typically comprises nine core courses, distributed evenly with three in each semester. The structure typically includes 4 hours of theoretical instruction and 3 hours of practical work, with three credits assigned to the theory and one credit to the practical component.

Level of Courses: 500 – 599

5.2. Skill-oriented Courses

The PG program with course work in the first 3 semesters and project work in 4th semester of 2nd year typically comprises six skill- oriented courses, distributed evenly with two in each of the three semesters. The structure typically includes three hours of theoretical instruction and three hours of practical work, with three credits assigned to the theory and one credit to the practical component.

Level of Courses: 400-499

5.3. Open Online Transdisciplinary Courses (OOTC)

The PG program typically comprises two open online transdisciplinary courses, distributed evenly with one in each of the first two semesters. Each online course is assigned two credits. A two-credit online course shall have a minimum duration of 40 hours of learning. A student can opt for two or three courses of shorter duration to acquire the assigned two credits. In the 1st semester, a student can opt for either the online course OR the equivalent two-credit course offered by the Yogi Vemana University, Kadapa. Open online transdisciplinary courses can be chosen based on the students' interest without confinement to traditional academic schedules.

Level of Courses: 300– 399

5.4. Courses on Indian Knowledge Systems (IKS)

The PG program typically comprises two IKS courses, distributed evenly with one in each of the first two semesters. The IKS course shall be an Audit Course, i.e., a mandatory course with only a Pass or Fail.



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In the 1st semester, a student can opt for either the online course OR the equivalent Two-credit IKS course offered by the Yogi Vemana University, Kadapa.

Level of Courses: 200– 299.

5.5. Open Online Skill Development Courses (OOSDC)

An 8-credit OOSDC is listed in Semester 4 of the 2-Year Master's Program with course work; students have the flexibility to pursue these courses at any point during their study period and fulfill the 8-credit requirement by the end of the 4th Semester. Each course is designed to be at least 20 hours in duration to earn ONE (01) credit. Students have the option to select one or two or more courses to accumulate the designated 8 credits.

Level of Courses: 300– 399

5.6. Project Work:

For the PG program with course work in the first 3 semesters, there shall be a research project work in the 4th semester of 2nd year. Though the research project needs to be completed in the 4th semester, the work could be initiated in the 3rd semester. The research project should be conducted with suitable experiments. At the end of the 4th semester, the student has to prepare and submit the dissertation in the standard format for evaluation.

The thesis should include (1) an Introduction of the research problem, (2) Review of the literature related to the problem, aims and objectives of the proposed research problem (3) Materials and Methods related to research problem (4) Results obtained from the methods using suitable statistical tools, (5) Discussion, Summary and Conclusion of the research findings on basis of results obtained and finally some of the significant references as a bibliography.

After evaluation, one copy of the thesis should be forwarded to the Controller of Examinations and another to the Departmental library.

Level of Course: 300 – 399

6. Evaluation of Student Performance

6.1. Assessment of Core and Skill Oriented Courses:

Summative and Formative Assessment

The recommended assessment distribution for the course entails a ratio of **60:40** between summative and formative assessments. In a total evaluation framework of 100



marks, 60% marks are allocated to summative semester end examinations, while there maintaining 40% marks are designated for formative internal assessments.

The Examination pattern for summative may be retained for summative assessment.

The proposed model for formative internal assessment is as follows:

Formative Internal Assessment (40 marks)

1. Mid-Semester Examinations (25 marks)
2. Attendance (5 marks)
3. Quiz (5 marks)
4. Student Seminar (5 marks)

The course shall comprise the first and second semesters during the 1st academic year and third and fourth semesters in the 2nd academic year. In the First, Second and Third semesters, there will be Five (05) theory papers (03 core + 02 skill orientation) and two practical's viz Practical-I consists of 03 core paper practical's; Practical-II consists of 02 Skill Orientation practical's) with Viva-Voce/Field Report /Records as the case may be. The maximum marks for each theory paper shall be 100 marks. Out of 100 marks in each theory paper, there will be 40 marks for the formative internal assessment and 60 marks for the summative semester-end examination. The total marks for four semesters put together shall be at least 2600, including Practical/Viva-Voce/Field Report/records/Project Report.

6.1.1. Mid-Semester Examinations (25 marks):

Two internal tests of one hour duration shall be conducted for 25 marks by the Department as per the academic schedule communicated by the Principal of the College. The better of the two tests shall be considered. A student absent for any test shall be deemed to have secured Zero (0) in that test, and no makeup test shall be conducted.

6.1.2. Attendance (5 marks)

Assignment of marks for attendance:

75%-80%	-1 mark
81%-85%	-2 marks
86%-90%	-3 marks
91%-95%	-4 marks
96%-100%	-5 marks



- a) A candidate shall maintain 75% attendance in each semester, including theory and Practical's etc.
- b) Condonation of shortage of attendance for theory classes alone may be recommended by the Coordinator/Head of the Department and Head of the Institution on medical grounds or other genuine reasons, provided the student puts up 62.5% of attendance throughout the semester. Candidates who put in 62.5% of attendance and apply for condonation must pay the condonation fee as prescribed by the university. In the case of a project, the student shall complete 75% of the attendance by attending extra classes on Sundays/preparatory holidays/vacation periods, etc, by paying an additional tuition fee, if necessary.
- c) A student who could not satisfy these requirements as given in 6.1.2 (a) and 6.1.2 (b) as the case may be shall have to take re-admission and repeat the semester and put in again minimum required attendance as stipulated in clauses 6.1.2 (a) and 6.1.2 (b) by paying additional tuition fee. The attendance put in by the candidate cannot be combined for computation.

6.13. Quiz (5 marks)

The Quiz must be done unit-wise. After the semester, the marks achieved for each unit will be averaged to contribute to 5 marks.

6.14. Student Seminar (5 marks)

The Seminar must be done unit-wise. After the semester, the marks achieved for each unit will be averaged to contribute to 5 marks.

6.2. Summative Examination

The university semester end examination with three hours duration shall be conducted for 60 marks. The setting of question papers shall be always by external examiners. All the semester's end theory and practical examinations conducted by the university shall be valued by both external and internal examiners. The project/dissertation submitted by the student at the end of 4th semester shall be valued for 350 Marks (200 Marks for dissertation, 50 Marks for presentation + 50 Marks for viva + 50 Marks for continuous internal assessment, i.e., regularity). The project shall be evaluated by a panel of two examiners comprising of faculty



supervisor as internal examiner, and one external examiner.

6.3. Assessment for Online Courses:

A student has to complete two Open Online Transdisciplinary Courses, two Indian Knowledge Systems, and one Open Online Skill Development Course in both years. If the online course is done from among the online courses offered in SWAYAM/ SWAYAM Plus/ UGC MOOCs/ NPTEL/IKS India/COURSERA/EDX Platform and any Government recognized platform, the credits and marks awarded shall be recognized and calculated for the SGPA and CGPA. The same shall be the case if any service provider conducts an online examination (proctored). If online examination is conducted either on SWAYAM/ SWAYAM Plus/ UGC MOOCs/NPTEL/IKS India/COURSERA/EDX platform or a proctored examination by the service provider, the University will conduct a pen and paper examination. The University will conduct examinations in the subsequent two semesters for the students who could not pass/appear in the end-term said online course exams (as per the UGC letter regarding: Framework for Universities to conduct examinations for SWAYAM courses (Dated on 27th August, 2024)).

6.4. Credit Transfer:

Prescribed courses taken by a student from YVU and its affiliated colleges and OOTC or IKS or OOSDC online courses from SWAYAM or UGC MOOCs or NPTEL or IKS India, or COURSERA and EDX Platform will be considered for credits at YVU.

Credits earned by students shall be deposited in the respective Academic Bank Account with ABC and shall be valid for seven years or as specified by the UGC from time to time for the purpose of commutation and award of any Degree or Diploma or Certificate: provided that once any credit is redeemed for the award of the aforementioned academic qualification, such credit shall be irrevocably debited from the respective student's Academic Bank Account. Once used, or redeemed, credits earned by a student cannot be re-used for the award of any other formal academic qualifications.

7. Promotions and Detention

A student shall be eligible for promotion to the next semester if he/she satisfies the minimum attendance requirements stipulated in regulations 6.1.2 (a) and 6.1.2 (b) in the semester



and register for the semester-end university examinations. Otherwise, the candidate will be detained. A candidate has to complete the course, including improvement, with internal assessment marks obtained during the study.

8. Qualifications for Pass

A candidate appearing for the first time for any semester-end university examination shall register for the whole examination. A candidate shall be declared to have passed the entire examination if

- He/she secures a minimum of 35% marks in each theory paper/practical semester-end university examination and an aggregate of 40% marks in each semester i.e., semester-end examinations and internal assessment marks put together. In case of project work, the candidate gets 35% marks in the dissertation and an aggregate of 40% marks in the dissertation and Viva-Voce put together. Paper-wise pass is allowed.
- The candidate shall be exempt in theory paper in which he/she secured a minimum of 35% of the marks in semester end examination and internal assessment marks, if the candidate secures overall aggregate of 40%.
- A candidate who could not secure the aggregate of 40% in any semester (i.e., semester end examination and internal assessment marks including seminar/record/viva-voce etc., put together) shall be declared to have been exempted in such theory paper in which he secured a minimum of 40% of the marks in semester end examinations and internal assessment marks put together.
- The failed candidates (supplementary) should secure not less than 35% marks in each theory paper/practical semester-end university examinations and 40% marks for passing the paper (s) (i.e., semester-end examinations and internal assessment marks put together) in which they have failed.
- A candidate who could not get the pass marks in the dissertation of the project work shall revise and resubmit the dissertation along with the next regular batch of students.
- A candidate who fails and wish to appear for the whole theory examination of any semester shall be allowed only after cancelling the results of the previous examination of that semester.



- g) Supplementary candidates shall register afresh and reappear for the paper in which he/she has failed in the examination; he/ she shall be declared to have passed the examination in such paper (s) when the candidate fulfills the requirements of the regulation 8(a), (b), (c) and (d).
- h) There shall be no separate supplementary examinations. The failed candidate shall register and reappear for the examination along with the next batch of regular candidates.
- i) The marks obtained by the candidates in the internal assessment/dissertation shall remain the same for improvement / failed candidates.

9. Letter Grades and Grade Points

The marks obtained in each paper by the candidate shall be given as letter grades and grade points on the 10-point scale, as shown below.

Range of marks	Letter Grade	Grade Points	Qualitative Meaning
90-100	O	10	Outstanding
80-89	A+	9	Excellent
70-79	A	8	Very Good
60-69	B+	7	Good
50-59	B	6	Above Average
45-49	C	5	Average
40-44	P	4	Pass
<40	F	0	Fail
0	AB	0	Absent

9.1. Computation of SGPA and CGPA

Students are classified on a 10-point scale based on their Semester Grade Point Average (SGPA) at the end of each semester and Cumulative Grade Point Average (CGPA) at the end of the course, as explained below.



a) Semester Grade Point Average (SGPA) is the ratio of the sum of the number of credits with the grade points scored by the student in all the courses taken by a student and the sum of the number of credits of all the courses taken by a student i.e.

$$SGPA (S_i) = \sum (C_i \times G_i) / \sum C_i$$

Where,

C_i is the number of credits of the i^{th} Course and G_i is the grade point scored by the student in the i^{th} course.

b) Cumulative Grade Point Average (CGPA) is also calculated in the same manner, taking into account all the courses undergone by a student over all the semesters of the programme i.e.

$$CGPA = \sum (C_i \times S_i) / \sum C_i$$

Where, S_i is the SGPA of the i^{th} Semester and C_i is the total number of credits in that semester.

The SGPA and CGPA shall be rounded to 2 decimal points and reported in the transcript.

$$\text{Aggregate Percentage} = CGPA \times 10$$

10. Award of Rank

There shall be no award of rank at the time of publication of results. However, the names of gold medal awardees, fellowship programmes, etc, shall be announced at the time of convocation only in their subject in the university based on total marks obtained in all the semesters put together and who passed all the semesters examinations in one attempt. Supplementary / improvement candidates are not eligible for the rank award.

11. Improvement

The candidates shall be permitted to improve their marks in one or more theory papers of any semester after completion of the course as per the regulations and syllabus in force. If the candidate improves, necessary corrections will be made to the marks statement and the provisional / degree certificate. The original marks remain unaltered if the candidate fails to improve their marks/grade. There shall be no provision for improving the marks obtained in internal assessment/records/practical/dissertations. **They shall not be eligible for the rank award.**

12. Conditions of Award of Degree

As per NEP 2020 guidelines, the MSc course comprises lateral entry and exit. On this



basis, the student shall be required to complete the degree;

- a) To have passed the examinations as prescribed above
- b) To have undergone the course of study in the university colleges over a period of two academic years, consisting of four semesters for a two-year M.Sc. which is equivalent to 84 credits with project work.
- c) To have lateral entry in the second year and undergo the course of study in the university colleges over two semesters for a one-year MSc program, which is equivalent to 40 credits with project work.
- d) When awarding the MSc degree for a one-year program with lateral entry, the provisional certificate and the conferred degree shall clearly state that the student has completed a one-year postgraduate program after finishing a four-year UG Honours program. Additionally, the number of credits earned by the student in the four-year UG program shall be noted.

13. Transitory Provision

- a) The M.Sc. course typically covers one or two academic years, comprising two or four semesters, respectively.
- b) A candidate can pursue two academic programs, M.Sc., in full-time physical mode (at YVU) and another in Open and Distance Learning (ODL)/Online mode (pursued with only such HEIs that are recognized by UGC/Statutory Council/Govt. of India for running such programs).

14. The university shall have the right to amend or modify any of the above regulations whenever necessary.



APPENDIX-I
MARKS DISTRIBUTION FOR THEORY AND PRACTICAL EXAMINATIONS

SEMESTER-I	Total Marks	Summative Examination (60 Marks)	Formative Internal Assessment (40 marks)			
		External	Internal	Attendance	Quiz	Students Seminar
CORE-1	100	60	25	5	5	5
CORE-2	100	60	25	5	5	5
CORE-3	100	60	25	5	5	5
SKILL-1	100	60	25	5	5	5
SKILL-2	100	60	25	5	5	5
CORE PRACTICAL-I (CORE 1+2+3)	150					
SKILL PRACTICAL-I (SKILL 1+2)	100					
TOTAL MARKS	750					
SEMESTER-II	Total Marks	Summative Examination (60 Marks)	Formative Internal Assessment (40 marks)			
		External	Internal	Attendance	Quiz	Students Seminar
CORE-4	100	60	25	5	5	5
CORE-5	100	60	25	5	5	5
CORE-6	100	60	25	5	5	5
SKILL-3	100	60	25	5	5	5
SKILL-4	100	60	25	5	5	5
CORE PRACTICAL-II (CORE IV+V+VI)	150					
SKILL PRACTICAL-II (SKILL 3+4)	100					
TOTAL MARKS	750					
		Summative	Formative Internal Assessment (40 marks)			



		Examination (60 Marks)				
SEMESTER-III	Total Marks	External	Internal	Attendance	Quiz	Students Seminar
CORE-7	100	60	25	5	5	5
CORE-8	100	60	25	5	5	5
CORE-9	100	60	25	5	5	5
SKILL-5	100	60	25	5	5	5
SKILL-6	100	60	25	5	5	5
CORE PRACTICAL-III (CORE 7+8+9)	150					
SKILL PRACTICAL-III (SKILL 5+6)	100					
TOTAL MARKS	750					
SEMESTER-IV	Total Marks					
PROJECT WORK	350	Dissertation 200	Presentation 50 +50+50 (Viva-Voce Continuous Internal Assessment)			
TOTAL MARKS	2600					

- Each CORE or SKILL Theory Paper consists of FOUR (4) Units. Theory examination is for 100 Marks (60 Marks external + 25 Marks internal examination + 05 Marks Attendance + 05 Marks Quiz + 05 Marks Students Seminar).
- Each Practical Paper is for 150 Marks. For the Semester End External Practical Examination, THREE CORE (03) Practical's (50 + 50 + 50) may be combined, and the examination will be conducted for 150 Marks for FOUR (04) hours. TWO Skill Oriented Courses (50 + 50) may be combined, and an examination will be conducted for 100 Marks for FOUR (04) Hours.



Semester- I

Core Course 1

ZOO CC-101: STRUCTURAL ANATOMY OF INVERTEBRATA & VERTEBRATA

Course outcome: It gives scope to learn, understand and appreciate diversity of anatomical features of invertebrates and vertebrates with respect to various physiological functions.

UNIT -I

- 1.1.Species concept (Typological species concept, Biological species concept, Evolutionary species concept, Polytypic & monotypic species, subspecies, infraspecific groups, super species, chrono species, ring species), International code of Zoological nomenclature- Operative principles, Taxonomical procedures, New trends in taxonomy
- 1.2. Organization of coelom and its significance: Acoelomata, Pseudocoelomata, Coelomata, Protostomia and Deuterostomia; Schizocoely and Enterocoely
- 1.3.Patterns of feeding and digestion in lower metazoans- Holozoic nutrition, Pinocytosis, Saprozoic nutrition, Myxotrophic nutrition, Nutrition of parasites.
- 1.4.Feeding in Polychaeta, Mollusca and Echinodermata

UNIT-II

- 2.1.Respiratory system in Invertebrates (Gills, Lungs and Trachea) and Mechanism of respiration
- 2.2.Circulatory system in Invertebrates (Annelids, Arthropods and Mollusca)
- 2.3.Advanced nervous system in Invertebrates (Annelida, Arthropoda and Mollusca)
- 2.4.Larval forms of Crustacea (Nauplius, Metanauplius, Protozoa, Zoaea, Cypris, Mysis, Megalopa, Phyllosoma, Alima) and larval forms of Echinodermata (Asteroidea Bipinnaria Larva, Ophiuroidea, Echinoidea, Holothuroidea, Crinoidea, *Doliolaria* Larva); Strategies and Evolutionary significance of larval forms.

UNIT-III

- 3.1.Vertebate integument and its derivatives: skin structure and functions, glands, scales, horns, claws, nails, hoofs, feathers and hairs
- 3.2.Evolution of heart in Vertebrate classes
- 3.3.Evolution of aortic arches in Vertebrates
- 3.4.Comparative account of respiratory organs in Vertebrates

UNIT-IV

- 4.1.Evolution of Urinogenital system in vertebrate series
- 4.2.Organs of Olfaction, taste and lateral line system
- 4.3.Comparative anatomy of the brain in relation to its functions
- 4.4.Spinal cord and cranial nerves in mammals

List of Practicals

- a) Museum study of all phylum wise representatives (Protozoa to Echinodermata)
- b) Museum study of all class wise representatives (Cyclostomes & Mammals)
- c) Techniques of collection and preservation with respect to insects and fishes
- d) Methods of collection, preservation and identification of fauna – zooplankton, insects,



fishes, birds

- e) Techniques of collection and preservation with respect to insects and fishes
- f) Dissection and observation of special structures a) poison gland of ants/spider; b) Mounting of mouth parts of mosquito/house fly c) Halteres of house fly/Honey bee
- g) Virtual dissection of weberian ossicle and brain in *Labeo rohita*
- h) Virtual dissection of cranial- Nerves of *Labeo rohita*
- i) Virtual dissection of cranial Nerves of frog/ toad
- j) Virtual dissection of circulatory (arterial & venous) system in *Calotes*
- k) Virtual dissection of Urinogenital system in *Calotes*

SUGGESTED READING MATERIAL

1. Hyman, L.H. The Invertebrates. Vol.1. Protozoa through Ctenophora, Mc Graw Hill Co., New York.
2. Barrington, E.J.W. Invertebrate structure and function. Thomas Nelson and Sons Ltd., London.
3. Jagerstein, G. Evolution of Metazoan life cycle, Academic Press, New York & London.
4. Hyman, L.H. The Invertebrates. Vol.8. Mc Graw Hall Co., New York and London.
5. Hyman, L.H. The Invertebrates. Vol.2 Mc Graw Hall Co., New York and London.
6. Barnes, R.D. Invertebrate Zoology, III edition. W.B. Saunders Co., Philadelphia.
7. Russel-Hunter, W.D. A biology of higher invertebrates, the Macmillan Co. Ltd., London.
8. Hyman, L.H. The Invertebrates smaller coelomate groups, Vol. V. Mc. Graw Hill, Co., New York
9. Read, C.P. Animal Parasitism. Prentice Hall Inc., New Jersey.
10. Sedwick, A. A student text book of Zoology, Vol. II and III. Central Book Depot, Allahabad.
11. Parker, T.J., Haswell, W.A. Text Book of Zoology, Macmillan Co., London.
12. Alexander, R.M. The Chordata. Cambridge University Press, London
13. Barrington, E.J.W. The Biology of Hemichordata and Protochordata. Oliver and Boyd, Edinburgh.
14. Bourne, G.H. The structure and functions of nervous tissue. Academic Press, New York.
15. Carter, G.S. Structure and habit in vertebrate evolution –Sedwick and Jackson, London.
16. Eccles, J.C. The understanding of the brain. McGraw Hill Co., New York and London.
17. Kingsley, J.S. Outlines of Comparative Autonomy of Vertebrates. Central Book Depot, Allahabad.



Core Course 2

ZOO CC-102: GENETICS & EVOLUTION

Course outcome: Students will learn about Mendelian and non-Mendelian inheritance. Understand the concepts multiples alleles, genetic disorders and karyotyping etc. Students acquire knowledge regarding different evolutionary theories and speciation.

UNIT – I

- 1.1.Principles of Mendelian Inheritance, extension and modifications to basic principles of heredity (Incomplete dominance, Co-dominance, Multiple alleles, ABO & Rh blood grouping) Inter allelic gene interaction (Epistasis, Complementary genes, Duplicate genes, Lethal genes), Identification of DNA as a genetic material.
- 1.2.Extra chromosomal inheritance; Linkage and crossing-over, chromosome mapping, tetrad analysis, uses of genetic mapping.
- 1.3.Microbial genetics: Methods of genetic transfers – transformation, conjugation, transduction and sexduction, mapping genes by interrupted mating, fine structure analysis of genes.
- 1.4.Mutations: a) spontaneous and b) induced mutations; c) Molecular basis of mutations

UNIT – II

- 2.1.Numerical and Structural abnormalities of human chromosomes and syndromes; Human karyotyping and human genome.
- 2.2.Pedigree analysis, Polygenic inheritance, Types of quantitative traits (continuous, meristic and threshold), Quantitative Trait Loci (QTL mapping).
- 2.3.Sex linked inheritance; Sex limited and sex influenced genes; Dosage compensation mechanism
- 2.4.Eugenics: Positive eugenics (Artificial insemination, sperm banks) and Negative eugenics (Amniocentesis, consanguinity), Genetic counselling

UNIT – III

- 3.1.Evolutionary time scale, Eras, Periods & Epoch - major events.
- 3.2.Theories of organic evolution- Emphasis on Darwinism and Lamarckism; Neo-Darwinism
- 3.3.Role of isolating mechanisms in evolution; Speciation and Models of speciation (Allopatric, sympatric and parapatric)
- 3.4.Genotype and allelic frequencies, Hardy-Weinberg equilibrium, non-random mating, consequences of homozygosity, factors affecting gene frequencies, heterosis.

UNIT – IV

- 4.1.Destabilizing forces (i) Natural selection (ii) Mutation (iii) Genetic drift
- 4.2.Phylogenetic gradualism & punctuated equilibrium
- 4.3.Patterns of Micro and Macro evolution
- 4.4.Gene Evolution and Amino acid sequence and phylogeny



List of Practicals

- Blood grouping and related problems
- Rh factor demonstration
- Multiple alleles and related problems
- Mendelian ratios and its related Problems
- Karyotyping
- Syndrome charts – demonstration
- Demonstration of Barr bodies
- Problems on Hardy Weinberg's law
- Pedigree Analysis, Polygenic inheritance,
- Multifactorial inheritance – Fingerprint analysis
- Trapping and Culturing native *Drosophila sp.* in Laboratory

SUGGESTED READING MATERIAL

- Genetics – Monrve W. Strickberger. 3rd Ed., May, 2000.
- Genetics – K.B. Ahluwallia – 1985.
- Principles of Genetics – E.J. Gardner. M.J. Simmons & D.P. Snustad.
- Molecular Biology of genes – Watson, J.D., N.H. Hopkins, J.W. Roberts, J.A. Steitz & A.M.
- Weiner. The Benjamin Cummings publishing company. Inc. Tokyo.
- Basic Human Genetics – E.J. Mange, Arthur P. Mange. Indian Print, 1997.
- Genetic disorders of Man by M.R. Goodman.
- An introduction to modern genetics by Ch. Waddingston.
- Dobzhansky, Th. Genetics and origin of species, Columbia University press.
- Dobzhansky, Th., F.J. Ayala, G.L. Stebbins and J.M. Valentine EVOLUTION: Surjeet publications, New Delhi latest edition.
- P.A. Moody Introduction to Evolution II ed/latest: Kalyani publishers, New Delhi.
- Hartl, D.L. A primer of population genetics, Sinauer Associatesm Inc., Massachusetts.
- Peter Volpe E. Understanding Evolution, University Book stall, New Delhi.
- An introduction to genetic analysis. Griffiths, A.J.F., J.H. Miller, D.T. Suzuki, R.C. Lewontin & W.M. Gelbark, W.H. Freeman and Company, New York.



Core Course 3

ZOO CC-103: METABOLIC REGULATION & CELL FUNCTION

Course outcome: It enables to understand primary aspects of carbohydrate, protein and lipid and nucleotide metabolism. It also highlights their interplay and biological significance in energy metabolism and cell function. It also helps to understand the mechanisms involved in catabolism of nitrogenous wastes and clinical disorders associated with nucleotide metabolism.

UNIT – I

- 1.1. Chemical bonds (Covalent, Hydrogen bonds, Ionic bonds, Van der Waals interactions)
- 1.2. Thermodynamic principles in biology
- 1.3. Outline classification of organic compounds (carbohydrates, proteins and lipids)
- 1.4. Orders of protein structure (primary, secondary, tertiary and quaternary)

UNIT – II

- 2.1. Glycolysis, TCA cycle and their biological importance
- 2.2. Pentose phosphate pathway, gluconeogenesis
- 2.3. Regulation of carbohydrate metabolism (Glycolysis and TCA cycle)
- 2.4. Mitochondrial electron transport system, Oxidative Phosphorylation

UNIT – III

- 3.1. Beta-oxidation of palmitic acid; Biosynthesis of long chain fatty acids (Palmitic acid)
- 3.2. Oxidative deamination, decarboxylation and transamination of amino acids.
- 3.3. Biosynthesis of Urea and detoxification of ammonia
- 3.4. Biosynthesis of polyamines

UNIT – IV

- 4.1. Nucleotides and types
- 4.2. Biosynthesis of Nucleotides
- 4.3. Degradation of Nucleotides
- 4.4. Clinical disorders of purine metabolism

List of Practicals

- a) Estimation of glucose
- b) Estimation of soluble and structural proteins
- c) Estimation of carbohydrates
- d) Estimation of amino acids
- e) Estimation of Lipids
- f) Estimation of Triglycerides

SUGGESTED READING MATERIAL

1. Robert K. Murrey, D.K. Granner, P.A. Mayes and V.W. Rodwell; Harper's Biochemistry McGraw Hill Publishers.
2. Biochemistry by A.L. Lehninger, Kalyani publishers, New Delhi.
3. D. Voet and J.G. Voet, Biochemistry, J. Wiley & Sons.



యోగి వేమన విశ్వవిద్యాలయం, కడప
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Re-accredited by NAAC 'A' Grade with CGPA of 3.13/4.0 (2nd Cycle)

4. David L. Nelson and Michael M. Cox, Lehninger; Principles of Biochemistry, McMillan worth Publications.



Skill Oriented Course 1

ZOO SOC-104: TOOLS AND TECHNIQUES IN BIOLOGY

Course outcome: Students gain knowledge about various tools and techniques used in biological systems and gives them insights about their usage in research and various diagnostics. Students will gain knowledge on animal cell culture techniques.

UNIT-I

- 1.1. Microscopy: Types of microscopes – Phase contrast microscope, Fluorescence microscope; Electron microscope – TEM and SEM
- 1.2. Centrifugation – basic principles, Types of rotors, high speed and ultracentrifuge
- 1.3. Principles of spectroscopy, Laws of Light absorption, applications of Colorimetry, Spectrophotometry
- 1.4. Measurement of pH and biological Buffers

UNIT-II

- 2.1. Chromatography – paper chromatography – thin layer chromatography
- 2.2. Ion exchange chromatography and affinity chromatography
- 2.3. Introduction to FPLC and HPLC
- 2.4. Radio isotope techniques – types of radio isotopes, detection and measurement of radioactivity, Applications of radio isotopes in biological sciences and safety measures

UNIT-III

- 3.1. Microtomy and staining procedures– types of microtomes, types of stains, staining procedures of biological materials
- 3.2. Electrophoresis: SDS-PAGE, Agarose gel electrophoresis
- 3.3. Blotting techniques
- 3.4. ELISA

UNIT-IV

- 4.1. Design and functioning of tissue culture laboratory methodology
- 4.2. Culture media preparation
- 4.3. Cell proliferation measurements
- 4.4. Cell viability testing and cell harvesting methods

List of Practicals

- a) Separation of biological compounds by paper chromatography
- b) Preparation of Buffers and measurement of pH
- c) Separation of biological compounds by TLC
- d) Absorption spectra of proteins and nucleotides
- e) Separation of mitochondria and differential centrifugation
- f) Separation of biomolecules using HPLC
- g) Preparation of cell culture media
- h) Separation of proteins by SDS-PAGE



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SUGGESTED READING MATERIAL (ALL LATEST EDITIONS)

1. Animal Cell Culture – A practical approach, Ed.John. R.W.Masters, IRL Press.
2. Introduction To Instrumental analysis, Ronert Braun. McGraw Hill International
3. A Biologists Guide to Principles and Techniques of Practical Biochemistry, K. Wilson & K.W. Goulding, ELBS Edn.
4. Advanced Micripipette Techniques for cell physiology. K.T. Brown and D.G. Flamming IBRO, Hand Book Series. A Wiley Interscience publications, John Wiley and Sons, New York
5. Neuro anatomical Techniques, N.J. Stransfed and T.A. Miller Springer Verlag, New YorkHeidelberg, Berlin, 1980.
6. Principles of Neuro psychopharmacology by Robert S. Feldman, Jerrold S. Meyer and Unida quenzer. Sinancer Associates Inc. Publishers. Sunderland, Massachusetts, 1997.
7. General Zoological Micro techniques – F.M. Weesner.



Skill Oriented Course 2

ZOO SOC-105: FIELD RESEARCH TECHNIQUES FOR WILDLIFE CONSERVATION

Course Objectives: To develop practical skills in field data collection, population monitoring, and habitat assessment.

- To introduce modern tools, methods, and ethical considerations in wildlife research.
- To build competence in data analysis, interpretation, and scientific reporting.

UNIT I:

- 1.1. Definition, scope, and objectives of field research in conservation.
- 1.2. Types of field research: observational vs. experimental.
- 1.3. Research design: formulating hypotheses.
- 1.4. Sampling methods, survey design.

UNIT II:

- 2.1. Methods for estimating wildlife population: total counts, sampling, mark-recapture.
- 2.2. Indirect methods: signs, tracks, scat analysis.
- 2.3. Ethics in wildlife research: permits, animal welfare, field safety.
- 2.4. Habitat mapping using GIS and Remote Sensing basics.

UNIT III:

- 3.1. Field equipment: GPS, camera traps, binoculars, range finders.
- 3.2. Capture, marking, and tagging techniques for different taxa.
- 3.3. Radio-telemetry and GPS tracking, Camera trapping,
- 3.4. Bioacoustics, drones in wildlife studies.

UNIT IV:

- 4.1. Field data management and quality control.
- 4.2. Basics of statistical analysis (intro to R, SPSS), Spatial data analysis using GIS.
- 4.3. Writing field reports, species accounts, and research papers.
- 4.4. New trends in wildlife field research.

Practical / Fieldwork:

1. Field trips to protected areas and wildlife reserves.
2. Hands-on training in survey methods, vegetation sampling, and species monitoring.
3. Demonstrations of telemetry, camera trapping, and habitat mapping.
4. Mini project: data collection, analysis, and presentation.

Suggested Readings

1. Wildlife Techniques Manual - Nova J. Silvy (Ed.)
2. Research Techniques in Animal Ecology - Boitani & Fuller
3. Measuring and Monitoring Biological Diversity - Heyer et al.
4. Recent articles from Journal of Wildlife Management and Conservation Biology.



Audit Course: Indian Knowledge Systems (IKS)

ZOO-IKS-107: INDIAN KNOWLEDGE SYSTEMS-AN OVERVIEW

UNIT -I: Bhāratīya Civilization and Knowledge Systems

- 1.1.Bharatavarsha- Genesis and Antiquity
- 1.2.The Vedic Corpus-Sruthis and Smrithis
- 1.3.The Vedic Corpus-Upavedas and Upanishads
- 1.4.Jainism and Buddhism

UNIT-II: Bharatiya Calendar

- 2.1.Different calendars of the World
- 2.2.Types of Indian Calendars
- 2.3.Astronomical Foundations of Calendars
- 2.4.Indian Festivals and Holidays

UNIT-III: Ancient Indian Philosophers

- 3.1.Ancient Philosophers: Agastya, Vedavyasa, Valmiki
- 3.2.Ancient Philosophers: Yājñavalkya, Gārgī
- 3.3.Vedanta School of Philosophy- Adi Sankaracharya
- 3.4.A glimpse of commentaries on Indian Philosophy by foreigners.

UNIT-IV: Contributions of IKS to Science and Technology

- 4.1.Ancient Scientists: Charaka and Suśruta, Vagbhata, Jivaka, Patanjali
- 4.2.Ancient Scientists: Vagbhata, Jivaka, Patanjali
- 4.3.Ancient Scientists: Nagarjuna, Kanada, Āryabhata,
- 4.4.Ancient Scientists: Varāhamihira, Baudhayana, Bhaskara-I&II

SELECTED REFERENCES

1. Barton, Gregory Allen (2002) *Empire Forestry and the Origins of Environmentalism*. CUP.
2. Bhag Chand Chauhan (2023) *Textbook on 'The Knowledge System of Bhārata'*
3. Pavan Kanwar & David Frawley (2011) *A Vedic Life: The practical Philosophy of Nature*.
4. Pradeep Kohle et al. (2006) *Pride of India-A Glimpse of India's Scientific Heritage*.
5. Pranay Lal (2016) *Indica: A Deep Natural History of the Indian Subcontinent*.
6. Rangarajan, M. (2001) *India's Wildlife History*
7. Sabareesh (2022) *A brief History of Science in India*.
8. <https://www.gyanvigyansangam.com>
9. Originofscience.com
10. Indianscriptures@gmail.com

CORE PRACTICAL-I (CORE 1+2+3)
SKILL PRACTICAL-I (SKILL 1+2)



Semester- II

Core Course 4

ZOO CC-201: MOLECULAR CELL BIOLOGY

Course outcome: Upon successful completion of this course, students will be able to understand and explain the structural and functional organization of cells at the molecular level. They will gain in-depth knowledge of the molecular mechanisms governing gene expression, signal transduction, cell cycle regulation, and programmed cell death. The course will also equip students with essential laboratory skills such as DNA/RNA isolation, PCR, protein analysis, and cell culture techniques.

Unit I

- 1.1. Overview of Cell Structure and Function: Prokaryotic vs. Eukaryotic cells, Cell theory and modern concepts.
- 1.2. Plasma Membrane-Structure and composition of biological membranes. Transport across membranes: passive, active, and facilitated.
- 1.3. Organelles and Their Functions: Endoplasmic reticulum, Golgi apparatus, Mitochondria and chloroplasts, Lysosomes, Peroxisomes, and Endosomes.
- 1.4. Cell Division and Regulation- Mitosis, Meiosis, Cell Cycle and its regulation.

Unit II

- 2.1. Central dogma of molecular biology. Mechanism of Semi-conservative replication. Enzymes involved in replication. Post replicational modifications and Inhibitors of replication. DNA damage and repair mechanisms.
- 2.2. Mechanism of Transcription- RNA polymerases and transcription factors. RNA Processing- mRNA capping, splicing, and polyadenylation.
- 2.3. Translation- Ribosomes, RNA Types, and translation process. Post-translational modifications.
- 2.4. Regulation of Gene Expression- General principles of gene regulation with reference to Lac and trp. Britten and Davidson model for Eukaryotic regulation.

Unit III

- 3.1. Cell signaling: Models of cell-cell signaling, steroid receptors, nitric oxide and carbon monoxide
- 3.2. Functions of cell surface receptors (G-protein coupled receptors, Tyrosine kinases, cytokine receptors, receptors linked to other enzymatic activities).
- 3.3. Pathways of intracellular signal transduction- cAMP pathways, cyclic cGMP.
- 3.4. Phospholipids and Ca^{2+} , Ras, Raf and MAP kinases.

Unit IV

- 4.1. Cell death and proliferation – Apoptosis: definition, morphological and biochemical differences between apoptosis and necrosis.
- 4.2. Mechanism (internal and external signals) and significance of apoptosis.
- 4.3. Hallmarks of cancer. Oncogenes and tumor suppressor genes.
- 4.4. Molecular mechanisms of transformation and metastasis.



List of Practicals

- Mitosis – Onion root tips
- Meiosis in flower buds/ Grasshopper testis
- Observation of prokaryotic and eukaryotic cells under microscope
- Giant chromosome in *Chironomus* Larva
- Effect of colchicine on mitosis
- Estimation of DNA by diphenylamine method
- Determination of purity and quantity of DNA
- Determination of melting temperature (T_m) of DNA
- Estimation of RNA by orcinol method
- DNA isolation from *E. coli*
- PAGE electrophoresis of proteins
- Problems related to molecular biology
- Southern and Western blotting
- Polymerase chain reaction

SUGGESTED READING MATERIAL

- Cell Biology (Fundamentals and Applications) By Gupta / Jangir, 2001; Agrobios, India.
- Cell and Molecular Biology by EDR De Robertis and EMR De Robertis Jr, Indian Edition, B.I. Publications, Pvt. Ltd.
- The Cell (A Molecular Approach) by Geoffrey M. Cooper, 2nd Edn. 2000, ISBN.
- Molecular Biology by David Freifelder, 1993
- Molecular Biology of Gene-by J.D.Watson, 1988
- Harper's review of Biochemistry by D.W. Martin et al 1990
- Biochemistry by A.L. Lehninger
- Cell and Molecular Biology-E.D.P. De Robertis and E.M.F. De Robertis
- Concepts in Molecular Biology-S.C. Rastogi, V.N. Sharma and AnandaTandon (1993)
- Genes VII by Benjamin Lewin
- Genes VIII and IX by Benjamin Lewin



Core Course 5

ZOO CC-202: DEVELOPMENTAL BIOLOGY

Course outcome: Students learn about germ cell determination, primordial germ cell migration, determination of germ layers in insects, nematodes, amphibians, reptilians, aves and mammals. They also learn about sex determination, oogenesis, fertilization and post fertilization events. Limb development, metamorphosis and morphogenesis also help in understanding of developmental biology very well.

UNIT I

- 1.1.Introduction development Biology: Scope and Science of developmental biology and its applications; Developmental patterns in metazoans.
- 1.2.Germ line determination: Germplasm and the determination of the primordial germcells.
- 1.3.Germ cell determination in Nematodes, Insects and Amphibians.
- 1.4.Germ cell migration in Drosophila Amphibians, Reptiles, Birds and Mammals

UNIT II

- 2.1.Gametogenesis: Morphological basis in animals, semen composition, formation, sperm function, Spermatogenesis
- 2.2.Leydig cells: Morphology, Differentiation, function and its regulation.
- 2.3.Oogenesis and Vitellogenesis: Ovulation, super ovulation and ovum transport in mammals.
- 2.4.Fertilization: Biochemistry of fertilization- Cell surface molecules in sperm-egg recognition in animals, mechanism of fertilization, molecular events during fertilization and post fertilization.

UNIT III

- 3.1.Creating Multi-cellularity: Cleavage types and patterns, comparative account of gastrulation, Neurulation
- 3.2.Germ layers: Ectoderm, Mesoderm and Endoderm
- 3.3.Programmed cell death: Incidence of apoptosis, apoptosis during animal development; Tetrapod limb development; apoptosis during limb development.
- 3.4.Metamorphosis in Insects and Amphibians

UNIT IV

- 4.1.Biology of sex determination: Testis determining genes, ovarian development, secondary sex determination in mammals, Environmental sex determination
- 4.2.Body axes: Establishment of body axes in mammals; Pattern formation in Drosophila, Amphibians and Chick.
- 4.3.Proximate tissue interaction: Competence and Induction
- 4.4.Genes and Morphogenesis

List of Practicals

- a) Observation of developmental stages in frog and chick
- b) Observation of different cleavage stages in the eggs of *Lymnea* (fresh water snail)



- c) Role of shell during developmental of chick
- d) Protein turnover during development of chick
- e) Phosphorous metabolism in developing chick embryo
- f) Calorific values during the development of chick
- g) Ontogeny of excretory pattern in developing chick
- h) Vitellogenesis in Crab
- i) Fecundity index in Crab
- j) Induced breeding in Frog
- k) Spermatozoa observation in different vertebrates

Reference Books:

1. Austen, C.R. and Short, R.V. Reproduction in Animals
2. Schatten and Schatten. Molecular Biology of Fertilization
3. F.T. Longo, Fertilization, Chapman & Hall
4. R.G. Edwards, Human Reproduction
5. S.F. Gillbert, Development Biology, Sinauer Associates Inc., Massachusetts
6. Ethan Bier the Coiled Spring Harlsor Laboratory Press, New York.



Core Course 6

ZOO CC-203: COMPARATIVE ANIMAL PHYSIOLOGY

Course outcome: Students will have scope to learn various concepts of physiological functions of the body such as digestion, respiration, excretion and the functions of nerves and muscles.

UNIT – I

- 1.1.Aim and scope of physiology; General physiological functions and principles.
- 1.2.Feeding mechanisms and regulation; Comparative physiology of digestion of Carbohydrates, Protein and Fats
- 1.3.Gastro-intestinal Hormones in regulation of digestion
- 1.4.Vitamins and their role in cellular metabolism

UNIT – II

- 2.1.Respiration- Types of Respiration, Respiratory organs, Mechanism of Respiration
- 2.2.Circulation of body fluids (Major types of body fluids, Blood, General properties of blood, Composition of blood, Blood groups and Transfusions) and types of hearts, heart beat and its regulation
- 2.3.Patterns of nitrogen excretion among different animal groups- Nitrogenous Waste Products, Morphology of the excretory system in different groups of animals, Mechanism of urine formation. Evolutionary significance;
- 2.4.Osmoregulation in different animal groups (aquatic and terrestrial): Biological importance of water, Body fluid compartmentation, Nature of the problem of osmoregulation in different environments; Invertebrate and Vertebrate body fluid regulation.

UNIT – III

- 3.1.Thermoregulation- Principles; Temperature as an environmental factor, Thermoregulation in Invertebrates and Vertebrates.
- 3.2. Temperature relations in Homeothermic animals and Poikilothermic animals; Physical heat regulation, Chemical heat regulation.
- 3.3. Adaptations to temperature extremes, Physiological basis of torpor, Hibernation and Aestivation.
- 3.4.Chronobiology- Clocks and rhythms, rhythm characteristics, Zeitgebers, Types of biological rhythms, Biological clocks and human health and diseases, Clock dysfunction and lifestyle related disorders, chronomedicine, chronotherapy.

UNIT – IV

- 4.1.Bioluminescence- Occurrence of bioluminescence among different animals, Biochemistry of luminescence, Control of bioluminescence, Functional significance of luminescence.
- 4.2.Chromatophores and their functional regulation; Mechanism of action of chromatophore, Movement of pigment, Control of Chromatophores, Factors controlling chromatophore system.
- 4.3.Muscle Physiology- Types of muscles, Structure and function of muscles, Theories of muscle contraction.
- 4.4.Physiology of receptors- Types of receptors- Photoreceptors, Mechanoreceptors and chemo



receptors and their functional significance.

List of Practicals

- Assay of salivary amylase
- Assay of lipase
- Assay of pepsin
- Assay of ascorbic acid
- Demonstration of cell fragility in different media (Iso, hypo and hyper)
- Muscle contraction demonstration
- Study of Myogenic and Neurogenic hearts
- Demonstration of rate of oxygen consumption in crab/fish.
- Estimation of hemoglobin and hematocrit in blood sample.
- Estimation of blood glucose (GOD-POD method)
- To study the bleeding time (Duke's Method) and clotting time of blood samples (Capillary method)
- Estimation of RBC and WBC count
- Detection of urea in urine sample

SUGGESTED READING MATERIAL

- C.L. Prosser. Comparative Animal Physiology. W.B. Saunders & Company
- R. Eckert. Animal Physiology. Mechanisms and Adaptation. W. H. Freeman & Company
- W.S. Hoar. General and Comparative Animal Physiology
- Schiemdt-Nielsen. Animal Physiology. Adaptation and Environment. Cambridge
- C.L. Prosser. Environment and Metabolic Physiology Wiley-Liss, New York.



Skill Oriented Course 3

ZOO SOC-204: TOXICOLOGY AND IMMUNOLOGY

Course outcome: Students get benefitted with the knowledge on various toxic substances and their effects on living organisms. Students get benefitted with the skill development in environment and occupational toxicology. They will gain practical experience in toxicity evaluation experiments. Imparts in depth knowledge of tissues, cells and molecules involved in host defense mechanisms. Students gain knowledge and understanding of types of immunity learn fundamentals of antigens, antibodies, complements and other immune components and its interactions. Understanding of immune mechanisms of tolerance, autoimmunity, Transplantation, vaccination.

UNIT-I

- 1.1.Origin and scope of Toxicology and Principles of Toxicology
- 1.2.Distribution, Excretion and Absorption of toxicants
- 1.3.Bio accumulation, bio-magnifications and Biotransformation of Toxicants
- 1.4.Teratogens and their effects on mammalian development

UNIT-II

- 2.1.Classification of pesticides, mechanisms of pesticide toxicity and detoxification mechanisms.
- 2.2.Toxicity Evaluation (LC50 & LD50) and factors affecting the Toxicity. Dosage forms and routes of drug administration
- 2.3.Toxic effects of metals (Cadmium, Lead and Mercury)
- 2.4.Sources of radiation, types of radiation and physiological hazards of radiation.

UNIT-III

- 3.1.Cells of the immune system: Lymphoid cells, Mononuclear cells, granulocytic cells, Mast cells
- 3.2.Organs of the immune system – primary and secondary lymphoid organs, lymphatic system
- 3.3.Antigens: Antigenic determinants or epitopes, immunogenicity, Haptens
- 3.4.Innate (Non-specific immunity): Anatomical barriers, phagocytosis, NK cells, Interferons

UNIT – IV

- 4.1.Humoral immunity: Immunoglobulins (fine structure of immunoglobulins and immunoglobulin classes); the complement system, Classical and alternate pathway, inflammation
- 4.2.Cell mediated immunity: Mechanism of cell mediated immunity; Brief account on Antigen presentation, Major histocompatibility complex
- 4.3.Antigen – Antibody interactions: Affinity, Avidity, Cross – reactivity, precipitation reactions, and Agglutination reactions.



4.4. Brief account on immunological disorders:

- Tolerance and Autoimmunity
- Transplantation
- Immunodeficiency diseases
- Immunization (Active and passive immunity)

List of Practicals:

- Determination of LC₅₀ and LD₅₀ of selected toxicants in different animals.
- Effect of temperature on the ciliary activity in the normal and pesticide/ metal exposed fresh water muscles
- SDH activity in activity in different tissues of frog/fish with reference to malathion/mercury/cadmium
- Effects of toxicants on the rate of oxygen consumption of aquatic animals
- Effects of toxicants on total proteins of fish/ frog
- Teratogenic effects of pesticides on mice
- Pesticidal effect on morphology of tissue using histopathological techniques
- Routes of drug administration
- Preparation of differential cell types (Macrophages, live cells, astrocytes)
- Double diffusion technique
- Radio immune diffusion
- ELISA Test Demonstration (Tridot)
- Agglutination test
- Rocket immunoelectrophoresis
- Immunoelectrophoresis demonstration

SUGGESTED READING MATERIAL

- Toxic interactions-R. S. Goldstein, W.R. Hewitt and J.B. Hook. Academic Press-1990
- Pesticides and human Welfare-D. L. Gunn and J.G.R. Stevens. Oxford University Press-1978
- Pesticides action and Metabolism-O' Brien
- Environmental toxicology of Pesticides-F. Mastimura, G. M. Boush and T. Misato
- The Encyclopedia of Americana-Vol.15
- Introduction to Biochemical Toxicology-E. Hodgson & F.E. Guthrie
- Casarett & Doull's -Toxicology-The basic science of poisons- C. D. Klassen, Mary, O. A & John Doull Kuby, J. (1998) Immunology, W.H. Freeman and Company, New York.
- Roitt, I., Brostoff, J. Male, D. (1999/2000) Immunology, 4th Edition. Harcourt Brace and Company Asia, Pte. Ltd., Singapore.
- Harpers Review of Biochemistry, Murray, Granier, Mayes and Rodwell, Lange Medical Publications, 25th Ed.
- Human Physiology by Stuart Era Fox, W.M.C. Brown Publishers, USA 1984 or Recent Edition.
- An introduction to Immunology by C.V. Rao, Narosa publishing house, 2002.
- Immunology introductory textbook by Nandini Shetty, Wiley Eastern Ltd.



Skill Oriented Course 4

ZOO SOC-205: BIOSTATISTICS AND BIOINFORMATICS

Course outcome: Students gain knowledge on scope and importance of biostatistics. Students learn and practice various statistical methods used in zoological studies and research. Learn basic applications of computers that are useful in biological applications. They also learn some of the basic bioinformatics tools used in retrieval, analysis of databases, sequence analysis.

UNIT – I

- 1.1. Definition - scope of biostatistics
- 1.2. Measures of central tendency – arithmetic mean, median and mode
- 1.3. Measures of dispersion -range, mean deviation, standard deviation, Standard error
- 1.4. Co-efficient of variation, types of correlation, linear regression analysis

UNIT –II

- 2.1. Concepts of probability, laws; Normal probability distribution and its application
- 2.2. Tests of significance: Students t-Test (simple, paired), F- test
- 2.3. Application of χ^2 (chi-square) test in biology and testing the goodness of fit.
- 2.4. Analysis of Variance (ANOVA), SPSS

UNIT – III

- 3.1. Operating system –Introduction–types of operating systems
- 3.2. MS – Office (ACCESS, EXCEL, WORD, POWER POINT), applications of computers in biology
- 3.3. Introduction to Artificial Intelligence, Machine Learning (ML) and Deep Learning (DL) in biology
- 3.4. Scope, importance and status of Bioinformatics

UNIT –IV

- 4.1. Biological data bases (Gene bank and Protein sequence database)
- 4.2. Sequence analysis: Pair wise and multiple sequence alignment
- 4.3. Introduction to genomics and proteomics and microarray analysis
- 4.4. Human genome project, Pharmacogenomics

List of Practicals:

- a) Problems related to Mean, mode and median
- b) Problems related to test of significance
- c) Analysis of variance (ANOVA)
- d) Regression curves
- e) Generation of graphs using MS-Excel
- f) Power point presentations
- g) Data analysis using SPSS
- h) Sequence data retrieval in FASTA format from NCBI database
- i) Searching with BLAST, Secondary structure Prediction
- j) Viewing of PDB files using Rosmol



k) Aligning sequence using Clustal X

SUGGESTED READING MATERIAL

1. Artificial Intelligence in Biological Sciences – [AbhayaBhardwaj](#) 2022 Microsoft Office, by Setultz, 1997.
2. Database processing by D.M. Kroenke, Galgotia Publications, 1990.
3. Introduction to Biostatistics – By Sokal – Rohlf (2nd Edn) freeman International Editor, 1971.
4. Bio – Statistics – An introductory text – Goldstein, A The Macmillan Co., NewYork, 1971.
5. Bio – Statistics - By Lewis Alvin E. Affiliated East – West press (P) Ltd., 1971.
6. Statistical analysis in Biology by Mather, K Chapman and Hall, London, 1972.
7. Probit analysis by finney, D.J.S. Chand & Co., Ltd., New Delhi
8. Biostatistics by Lewis Alvin (1971) Affiliates East West Press Pvt., Ltd., New Delhi.
9. Statistical methods in Biology by Bailey Norman T.J. (1965) The English LanguageBook Society & the English university Press Ltd.
10. Bioinformatics. Murthy, C. S. V. Himalaya Publishing House, Hyderabad
11. Bioinformatics by Andreas D. Baxevanis and B.F. Francis Ouellette, 2nd Ed., 2002.
12. Basic Bioinformatics by S. Ignaeimuthi, S. J. Narosa publications, 2005
13. Introduction to Bioinformatics, S. Sundara Rajan and R. Balaji, Himalaya PublishingHouse, 2003.



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Audit Course: Indian Knowledge System 2

ZOO-IKS-207: CONTRIBUTIONS OF IKS TO SCIENCE & TECHNOLOGY

UNIT-I: Introductory Concepts

- 1.1. Concept and scope of Ancient Indian Science & Technology
- 1.2. Concept of Living and non-living (panchabhuthas)
- 1.3. Foods of India.
- 1.4. Concept of Universe, Life, Gravity and Laws of Motion, Mathematics, Geometry, Astronomy, Aeronautics, Metallurgy

UNIT-II: Forestry in Ancient India

- 2.1. Aranya and Vana in India
- 2.2. Aranyaka and Upanishad 'Bṛhadaranyaka'
- 2.3. Ancient Forests of India- a general account
- 2.4. Conservation of forests and biodiversity in ancient Indian culture

UNIT-III: Medicinal Knowledge Systems in Ancient India

- 3.1. Ethnobotany and Ethnology of India
- 3.2. Ayurveda and Siddha Medicine. Significant medicinal plants
- 3.3. Surgery and cloning technologies
- 3.4. Integrated Approach to Healthcare

UNIT-IV: Miscellaneous Concepts

- 4.1. Vedic Taxonomy, Anatomy, Pathology and Physiology
- 4.2. Agriculture in Ancient India
- 4.3. Ecology, Hydrology and Water harvesting
- 4.4. Sacred Groves and Landscapes. Sacred Plants and animals

SELECTED REFERENCES:

1. Barton, Gregory Allen (2002) Empire Forestry and the Origins of Environmentalism. CUP.
2. Bhag Chand Chauhan (2023) Textbook on 'The Knowledge System of Bhārata'.
3. Pavan Kanwar & David Frawley (2011) A Vedic Life: The practical Philosophy of Nature.
4. Pradeep Kohle et al. (2006) Pride of India-A Glimpse of India's Scientific Heritage.
5. Pranay Lal (2016) Indica: A Deep Natural History of the Indian Subcontinent.
6. Rangarajan, M. (2001) India's Wildlife History
7. Sabareesh (2022) A brief History of Science in India.
8. <https://www.gyanvigyansangam.com>
9. Originofscience.com

CORE PRACTICAL-II (CORE 4+5+6)

SKILL PRACTICAL-II (SKILL 3+4+ Zoological study tour - 30 marks)



Semester -III
Core Course 7
ZOO CC-301: NEUROBIOLOGY

Course outcome: As neurobiology segment covers basic aspects of types and structure of neurons, mechanism of signal transmission along nerves. It helps students to comprehend nervous system function. Further it also explains in detail about types of neurotransmitters, their production, storage, release and inactivation to understand their precise role in neuronal function.

UNIT: I

- 1.1.Introduction to Nervous system
- 1.2.Organization of Nervous system; Central nervous system (Forebrain, Midbrain and Hind brain) Peripheral nervous system-
- 1.3.Spinal cord organization- Spinal nerves, Cranial nerves
- 1.4.Types of nerve cells - Functional neurons and Glial cells

UNIT-1I

- 2.1.Micro anatomy of neurons.
- 2.2.Autonomic nervous system.
- 2.3.Bioelectrical properties of neurons (Resting membrane potential- Nernst equation; Sodium and potassium pump; Propagation of nerve impulse, generation of the action potentials
- 2.4.Synapses: Structure and Integration (Types of synapses; Ultra structure of synapse Chemical transmission; Electrical transmission)

UNIT-II

- 3.1.Chemical composition of the nervous system-cerebrospinal fluid-CNS barriers
- 3.2.Synthesis – storage-release and inactivation mechanisms and functions of the following neurotransmitters; Acetylcholine & Catecholamines (Nor epinephrine, Epinephrine, Dopamine)
- 3.3.Amino acid Neurotransmitters-Glutamate and GABA
- 3.4.Neuropeptides (Oxytocin and Vasopressin)

UNIT-IV

- 4.1.Functional disorders- Headaches, Epilepsy, Neuralgia
- 4.2.Neuro degenerative disorders-Alzheimer's disease, Parkinson's disease and multiple Sclerosis
- 4.3.Neuromuscular disorders-Muscular dystrophy and Myasthenia gravis
- 4.4.Peripheral nerve disorders - Peripheral neuropathy and Carpal tunnel syndrome

Practicals in Neurobiology

- a) Heteropolar and multipolar neuron
- b) Sensory neurons
- c) Coelenterata nerve net



- d) Pyramidal cell from cortex
- e) Motor neuron from spinal cord
- f) C.S. of spinal cord
- g) Bipolar cell from olfactory bulb
- h) Neuromuscular junction
- i) Stretch receptors in Cray fish
- j) Organization of *Sepia* central nervous system
- k) Synapse enlarge
- l) Stellate ganglion in *Sepia* isolation
- m) Identification of different regions of mice brain
- n) Spinal reflexes in decerebrated frog

SUGGESTED READING MATERIAL

1. Neurobiology. Shepherd, G.M. Oxford University press, London.
2. Basic Neurochemistry-G.J. Siegal, R.W. Albers, B.W. Agranoff, R. Katzman (1981) Little, Brown and company. Boston.
3. Introduction to Nervous system-T.H. Bullock, R. Cork, A. Graner (1977); W.H Free-man & Co.
4. Principles of Neural Science –E.R. Kandel and J.H. Schwartz. (1981); Elsevier/North Holland. NY. Oxford.
5. Mechanism of Drug Action on the Nervous System- M.A.B. Brazil, R.W. Ryall. (1979); Cambridge University Press. Cambridge, London and New York.
6. The Bio Chemical basis of Neuropharmacology-J.R. Cooper, F.E. Bloom, &R.H. Roth. (1982); Oxford University Press, NY and London.



Core Course 8

ZOO CC-302: APPLIED ANIMAL BIOTECHNOLOGY

Course outcome: Gives knowledge to culture of animal cells and its culture medium. Learn basic concepts and principles of recombinant DNA technology, Gene manipulation for transgenic animal production and therapeutics/vaccine production. Provides knowledge on Livestock, improvement aquaculture, generation of Chimeras. Provides knowledge on Intellectual property rights and genetically modified organisms.

UNIT-I

- 1.1.General Introduction and Achievements of Biotechnology
- 1.2.Enzymes used in gene cloning - Restriction endonucleases, DNA ligases, Kinase, Phosphatase, Nucleases, Polymerases, Reverse transcriptase
- 1.3.Cloning vectors (Plasmids, Phages, Cosmids, yeast Shuttle vectors), viral vectors (SV40, Adenovirus and Baculovirus) used in Gene cloning.
- 1.4.Cloning and selection strategies of recombinants (antibiotic selection, blue white screening, colony hybridization, Fluorescence in-Situ Hybridization (FISH) and immunological test.

UNIT-II

- 2.1.Preparation of cell lines, types of cell lines. Types of Stem Cells, Stem Cell Therapy
- 2.2.Applications of cell culture in Veterinary– Disease diagnosis, virus vaccines, hormones
- 2.3.Application of Biotechnology in Medicine- Production of monoclonal antibodies (Hybridoma technology), Production of vaccines and Production of Growth Hormone
- 2.4.Gene therapy: Introduction, principle of gene transfer and examples (Adenosine deaminase deficiency disease, Duchenne Muscular dystrophy disease and Cystic fibrosis)

UNIT-III

- 3.1.Livestock improvement: Manipulation of reproduction in animals (Artificial insemination, multiple ovulations, *in vitro* fertilization, Embryo transfer technology)
- 3.2.Methods of gene transfer – Microinjection, electroporation, lipofection and viral mediated gene transfer techniques
- 3.3.Generation of chimeric, transgenic and knockout mice and other animals and their characterization. Gene editing- Gene silencing-CRISPR-associated protein-9 nuclease (Cas9) technology
- 3.4.Potential application of transgenic animals: models for various diseases/disorders, production of peptides and proteins of biopharmaceutical interest (molecular farming)

UNIT-IV

- 4.1.Growth hormone transgenics and stem cell technology for betterment of aquaculture. Sex reversal in fishes and their applications, Production of mono sex populations. Aquaculture and fish seed production: Hypophysation, hCG injections
- 4.2.Marine bio/fish resources and its applications in pharmaceutical and Nutraceutical Industries
- 4.3.Intellectual Property Rights: Introduction; Types of IP; Patents and its types.



4.4.Trademarks, Copyright & Related Rights, Protection of GMOs; ethical and legal issues in Biotechnology.

List of Practicals:

- Instrumentation in animal biotechnology laboratory
- Preparation of different types of culture media
- Isolation of genomic DNA
- Bacterial Plasmid DNA Isolation
- Restriction digestion and ligation of vector and insert gene
- Bacterial transformation using plasmid
- Cell counting using hemocytometer
- Staining and viability testing of animal cells
- Media preparation and membrane filtration
- Preparation of single cell suspensions from spleen and thymus
- Sterilization techniques (Physical and chemical)

SUGGESTED READING MATERIAL

- Animal Biotechnology-M.M. Ranga, Agrobios (India)-2000
- Biotechnology-Fundamentals & Applications- S. S. Purohit & S. K. Mathur, Agro Botonics-1999
- A text book on Biotechnology-(II Ed.) H.D. Kumar. EWP-Private Ltd., New Delhi- 1998
- A text book of Biotechnology-R.C. Dubey. S. Chand & Company Ltd., New Delhi-1996
- Biotechnology-V. Kumaresan. Saras Publication-1994
- Animal Biotechnology – Recent concepts and developments. P. Ramadass, MJP Publications, Chennai, 2009
- Venkitaraman: Economic Zoology (Sudarsana Publishers, 1983)
- Srivastava: A text book of Applied Entomology, Vol. II and III (Kalyani Publishers, 1988, 1991)
- Shukla & Upadhyaya: Economic Zoology (Rastogi Publishers, 1999-2000)
- Dunham R.A. Aquaculture and Fisheries Biotechnology and Genetic Approaches. CABI Publishing, UK.
- Animal Transgenesis and Cloning. Edited by L. M. Houdebine, Wiley, USA.



Core Course 9

ZOO CC-303: ECOLOGY AND ECONOMIC ZOOLOGY

Ecology Course outcome: Students learn about different types of Ecosystems, energy Flow patterns in different ecosystems. They also learn about, Air & water pollution Parameters in India.

Economic Zoology Course outcome: Gives an outlook on characterization of Invertebrate and vertebrate phyla. Explain the role of various animals of economic importance. Gaining knowledge on the culture beneficial animals like earthworm, honeybees, silkworms, poultry and dairy. Gain knowledge on diseases causing organisms and its preventive mechanism.

UNIT- I

- 1.1.A general account on ecosystem and their environments
- 1.2.Freshwater: Classification and Characteristics, Eutrophication, seasonal changes
- 1.3.Marine: Classification and Characteristics
- 1.4.Terrestrial: Characterization of Forests-Grass lands-Tundra-Desert

UNIT- II

- 2.1.Dynamic view of ecosystem
- 2.2.Energy Flow patterns in different ecosystems
- 2.3.Air Pollution: Criteria and Standards in India, Health hazards and Toxicology –Green House gases and Green House effect
- 2.4.Water Pollution: Criteria and Standards in India, Health hazards

UNIT- III

- 3.1.Importance of Economic Zoology in sustainable development
- 3.2.Beneficial animals: Corals-Earthworm -Vermiculture and Vermicompost technologies
- 3.3.Beneficial Insects (Apiculture, Lac culture, Sericulture)-Economic importance.
- 3.4.Aquarium Fish Keeping, Exotic and Endemic species of Aquarium Fishes; Biology of Aquarium Fishes, Aquarium Maintenance and Live fish transport

UNIT-IV

- 4.1.Pearl and mollusc culture
- 4.2.Poultry keeping: Morphology of different breeds of Chicken-Brooding and Rearing of Chicks-Processing of Egg, Meat and By-Products of Poultry.
- 4.3.Dairy farm management, Milch breeds. Draught breeds, Dual purpose breeds and New Cross breeds of Cows and Buffaloes in India.
- 4.4.Zoonotic diseases: Impact on Human health

List of Practicals:

- a) Estimation of dissolved oxygen content in different water samples
- b) Effect of Photo periodism on CO₂ levels in different water samples
- c) Estimation of Organic matter in water and soil samples



- d) Different animals of the globe- endemic to parts of world
- e) Study of some specimens of beneficial insects
- f) List out beneficial animals
- g) List out Zoonotic diseases and their causative agents
- h) Visit to nearby Zoo, Butterfly Park, Insect museum at Yogi Vemana University, and submit the reports as a part of practical work

SUGGESTED READING MATERIAL

1. Environmental Physiology of desert organism. Ed. by N.F. Hadley, Dowden Hutchinson and Ross, Inc. Penn. USA.
2. The Ecology of waste water treatment- H.A. Hawkes pergoman press, 1963.
3. Biochemical ecology and water pollution-P.R. Dugan, plenum press, London, 1972.
4. Ecology & Environment- P.D. Sharma, 1991.
5. Field Biology & Ecology- Allen H. Benton & E. Werner, JR, 1980.
6. Encyclopedia of environmental pollution and control, enviromedia, Karad, Vol.1&2, R. K. Trivedy.
7. Eco technology for pollution control and environmental management, Enviromedia, Karad, R.K. Trivedy.
8. Health hazards and human environment, World Health Organization (WHO) 1972.
9. Current pollution researches in India - R. K. Trivedy and P. K. Goel. Karad.
10. Chemical and biological methods for water pollution studies-R.K. Trivedy and P. K. Goel, 1984.
11. Animal Physiology - Adaptation & Environment, 4th Edition-Knut Schmidt-Nielsen-Cambridge University Press.
12. Environmental Biology and Toxicology P.D. Sharma, Rastogi Publications, Meerut (India), 1998.
13. Biodegradation & Bioremediation-2nd editon, Martein Alexander-Academic Press, 1999 USA.
14. Water Treatment and purification technology - W.J. Ryan, Agro bios (India), Jodhpur, 2002.
15. Methods in Environmental Analysis - Water soil and air by P.K. Gupta - Agro bios (India), Jodhpur, 2001.
16. Hyman, L.H. The Invertebrates. Vol.8. Mc Graw Hall Co., New York and London
17. Hyman, L.H. The Invertebrates. Vol.2 Mc Graw Hall Co., New York and London
18. Kingsley, J.S. Outlines of Comparative Autonomy of Vertebrates. Central Book Depot, Allahabad
19. Economic Zoology, Dr. G.S. Sukla and Dr. V.B. Upadhyay, Rastogi Publications, 5th Edition, 2013
20. Economic Zoology by Venkitaraman PR, Publication: Kottayam V. Publishers 1989
21. Text Book of Economic Zoology by Venkitaraman, P R, 1983



Skill Oriented Course 5

ZOO SOC-304: AQUACULTURE AND HEALTH MANAGEMENT

Course outcome: The course imparts knowledge and expertise to establish aquaculture ponds and maintenance of physico-chemical parameters for optimal growth of cultured species. It also helps gain knowledge on nutritional requirements of fish/prawn and stocking management. They also gain practical experience in fish pathology and disease management. Students will get a unique opportunity to get associated with various aqua industries and achieve employment.

Unit-I: Aquaculture systems

- 1.1.Scope and origin of aquaculture; different farming systems and methods; warm water and cold water aquaculture; sewage – fed fish culture.
- 1.2.Site-selection for pond; types of ponds; sources of water; losses of water; water budgets of embankment ponds and an excavated pond; water exchange; design and constructions of pond/aquaculture farms and hatcheries; aeration and aerators.
- 1.3.Selection of aquaculture species; Pre-stocking and stocking Management; species combinations; stocking density.
- 1.4.Post stocking Management; length-weight relationship (LWR) and Condition factor; methods of determination of age in fishes and shellfish based on length data.

Unit-II: Nutrition and feeding

- 2.1.Nutritive needs of finfish (carps, shrimps and prawns); Natural feed and live feed.
- 2.2.Types of food; Types of feeding habits; Food and Feeding Habits of prawns, shrimps, crabs and brackish water fishes; feeding strategies
- 2.3.Supplementary feeds and formulations; types of feed; feed consumption, nutrient source and feed ingredients
- 2.4.Feeding and shrimp production; water quality; feeding rates; feeding devices; Feeding methods; relationship between feeding and growth; Feed ratio and feeding schedule; Check tray monitoring; Feed management in shrimp (*Penaeus monodon* and *Litopenaeus vannamei*)

Unit-III: Freshwater and coastal aquaculture, Fish pathology and Disease management

- 3.1.Culture of Indian Major Carps: exotic carps of Fish Hatcheries and their management. Culture of Marine/ brackish water fish (*Lates calcarifer* and *Etroplus suratensis*) and mud crabs (*Scylla serrata*).
- 3.2.Culture of air breathing fishes and ornamental fishes; Culture of freshwater and marine prawn/shrimp culture; Pearl culture (marine and freshwater) and Types of pearls.
- 3.3.Viral, Bacterial, Fungal protozoan, helminthic and crustacean diseases in fishes, prawn and shrimps; Nutritional deficiency diseases; environmental stress diseases.
- 3.4.Disease diagnostic methods; Methods for disease control and management. Role of chemotherapeutic agents, Vaccines, adjuvants, probiotics and chemicals in diseases prevention.

Unit-IV: Fish processing Technology

- 4.1.Biochemical composition of raw fish and shellfish; Nutritive and medicinal value of fish,



Nutritional value of preserved and processed fish.

- 4.2. Fish Decomposition; decay and spoilage of fish; effect of temperature on spoilage; Intoxications and food poisoning from fish.
- 4.3. Principles of fish preservation; Methods of preservation; industrial methods of freezing fish and shellfish; problems in fish preservations; fish preservatives.
- 4.4. Fish Byproducts- Methods of extraction of fish liver oil; processing of Fish meal, Fish Silage, fish sausages, Fish manure and guano; Chitosan, chitin pearl essence, Isinglass, Fish leather, Fish Caviar, Fish Macaroni.

List of Practicals:

- a) Identification of commercially important finfish-marine fishes, freshwater fishes, exotic carps, ornamental fishes, predatory fishes and weeds fishes.
- b) Identification of commercially important shellfish-crabs prawns and shrimps.
- c) Identification of fins and scales of fish; mounting of scales.
- d) Estimation of organic matter and Biological oxygen demand (BOD).
- e) Effect of eye stalk ablation on oxygen consumption in crab.
- f) Collection and preservation of pituitary gland in fish.
- g) Proximate composition of aquaculture feeds – Proteins, carbohydrates, lipids, moisture, and ash content.
- h) Determination of amino acids profiles through paper chromatography.
- i) Isolation and characterization of intestinal pathogens.
- j) Identification of disease causing microbes and identification of various diseases in finfish and shell fish.
- k) Host examination – collection of parasites; histopathology of organs of diseased fish; Slides of fish parasites.
- l) Determination of Length-weight relationship and condition factor of fish.
- m) Method of evaluation of freshness of fish.
- n) One day field trip to Hatcheries and Aqua farms (10 marks)

Field visit to hatcheries: Hatcheries visits will allow students to learn physico-chemical parameters for optimal growth of cultured species. Students will gain knowledge on nutritional requirements of fish/prawn and stocking management pathology and disease management. Students will get a unique opportunity to get associated with various aqua industries and achieve employment.

SUGGESTED READING MATERIALS

1. Pillay, T.V.R. 1993. Aquaculture: Principles and practices. Fishing new books. Black well Scientific Publications.
2. MPEDA, 1991. Handbook on shrimp farming, Kochi, India.
3. Jhingran, V.G. 1982. Fish and fisheries of India. Hindustan Publishing Corporation, New Delhi.
4. MPEDA, 1990. Handbook on Aqua farming-Live feed.
5. MPEDA, 1990. Handbook on Aqua farming-Aquaculture feed.
6. MPEDA, 1990. Aquaculture engineering and water quality management. Cochin, India.



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7. Rath, R. K, 2000. Freshwater Aquaculture. Scientific Publishers (India), Jodhpur.
8. Piska, R. S. 1999. Fisheries and Aquaculture. Lahari Publications.
9. Pandian, T.J. 2001. Sustainable Indian Fisheries. National Academy of Agricultural Sciences.
10. Kabata, Z. 1985. Parasites and Diseases of fish cultures in the Tropics. Taylor and Francis, London.
11. Das, M. k. and Das, R. K. 1997. Fish and prawn diseases, Inland Fisheries Society of India, Barrackpore.
12. Chakrabarthy, N.M. 1995. Disease of cultivable fresh water fishes and their control. Daya Publishing house, New Delhi.
13. Ellis A. E. (Edr.) 1995. Fish and shell fish pathology. Daya publishing houses, New Delhi



Skill Oriented Course 6

ZOO SOC-305: ENDOCRINOLOGY & ENZYMOLOGY

Course outcome: Students gain basic knowledge about various hormones from endocrine glands. Students will learn the concepts of mechanisms of hormone action at cellular and molecular level, their biosynthesis, their role in metabolism and homeostasis and further their role as drugs. Similarly, students can also learn about enzymes, its isolation, purification and distribution; their mechanism and regulation. They gain knowledge about the process of enzyme engineering, immobilization and their applications.

UNIT-I

- 1.1.Introduction to Endocrinology- Historical back ground, characteristic features of hormones
- 1.2.Classification and chemical nature of hormones; Mechanism of hormone action
- 1.3.General account of Semiochemicals, Types of semiochemicals - Pheromones and allelomones; Role of semiochemicals in Integrated pest management
- 1.4.Hormones in Crustaceans and Insects - growth, development and reproduction.

UNIT-II

- 2.1.Structure and functions of hormones of Major endocrine glands (Pineal, Pituitary, thyroid, Parathyroid, Pancreas, Adrenal)
- 2.2.Hormones in female sexual cycle, Pregnancy and lactation; Hormones of Testis and regulation of spermatogenesis
- 2.3.Hormones and homeostasis (Glucose, Calcium, Phosphate, Water); Hormonal regulation of carbohydrate, nitrogen and lipid metabolism
- 2.4.Hormones and human health; Hormones as pharmaceuticals

UNIT-III

- 3.1.Classification of enzymes and nomenclature; Enzyme specificity, Intracellular distribution and isolation of enzymes, Quantitative measurement of enzyme activity
- 3.2.Bioenergetics- Kinetic theory and collision theory; Mechanism of enzyme action
- 3.3.Effect of enzyme concentration, reactant concentration, substrate concentration, pH and temperature
- 3.4.Determination of kinetic constants and Inhibition of enzyme activity

UNIT-IV

- 4.1.Kinetics of Allosteric enzymes; Regulation of enzyme activity; Catalytic efficiency of enzymes
- 4.2.Immobilized enzymes and their applications
- 4.3.Isozymes and Enzyme engineering
- 4.4.Enzymes in clinical diagnosis

List of Practicals:

- a) Observation of the histological section of the pituitary, adrenals, pancreas and gonads
- b) Isolation and extraction of pituitary gland from fish



- c) Estimation of glucose levels in the blood of frog/rat exposed to adrenaline and insulin
- d) Estimation proteins in the reproductive tissues of a fish injected with pituitary extract
- e) Estimation of SDH activity in the haemolymph of eyestalk ablated crab
- f) Estimation of oxygen consumption in eyestalk ablated crab
- g) Demonstration on the effect of ligature on the development of larvae of insects
- h) Estimation of glucose in alloxan-induced diabetes
- i) Effect of adrenalectomy on total proteins in the liver of albino rats
- j) Diagnosis of pregnancy by the presence of HCG in urine using hCG strip test.
- k) Effect of pH on SDH activity and determination of ionizable groups
- l) Effect of temperature on SDH activity
- m) Determination of kinetic constants such as K_m and V_{max}
- n) Inhibitor sensitivity (determination of IC_{50})
- o) Effect of inhibitors on SDH activity and determination of inhibitors constant
- p) Estimation of GOT and GPT in the serum samples
- q) Isolation and purification of arginase
- r) Isolation of LDH isozymes using electrophoresis
- s) Determination of K_s (substrate constant) for any allosteric enzyme using Hill equation
- t) Characterization of any selected enzyme using bioinformatic tools (sequence determination, homology search, structure, genomic sequence, cDNA sequence, dendrogram, metabolic function.

SUGGESTED READING MATERIAL

1. Barrington. E.J.W. General and comparative Endocrinology Cambridge Press, Oxford.
2. Bentley, P.J. Comparative Vertebrate Endocrinology, Cambridge Press, Oxford
3. Williams, R.H. Text Book of Endocrinology, W.B. Saunders Co., Philadelphia.
4. Martin, C.R. Endocrine Physiology. Oxford Univ. Press, Oxford.
5. Prakash S. Lohar. Endocrinology-Hormones and human health-2005. MJPPublishers-Chennai

CORE PRACTICAL-III (CORE 7+8+9)

SKILL PRACTICAL-III (SKILL 5+6)



యోగి వేమన విశ్వవిద్యాలయం, కడప
YOGI VEMANA UNIVERSITY



Re-accredited by NAAC 'A' Grade with CGPA of 3.13/4.0 (2nd Cycle)

MODEL QUESTION PAPER
Yogi Vemana University - Kadapa
M.Sc., Zoology I, II, III SEMESTERS
Title of the Paper: XXXXX
QP Code: XXX
(w.e.f. Academic Year 2025-2026)
(No additional sheet will be supplied)

Time: 3 hours

Max. Marks: 60

PART – A (Marks: 4x3= 12 Marks)

(Two questions from each Unit Answer any **FOUR** questions.

Each answer should not exceed ONE (1) Page

All questions carry equal marks

1. From **UNIT - I**
2. From **UNIT - I**
3. From **UNIT - II**
4. From **UNIT - II**
5. From **UNIT - III**
6. From **UNIT - III**
7. From **UNIT - IV**
8. From **UNIT - IV**

PART B (Marks: 4 x 12 = 48)

Answer **ALL** questions

Two questions from each Unit with internal choice, answer should not exceed SIX (6) Pages

All questions carry equal marks

- | | |
|------|-------------------|
| 9a. | Unit – I |
| | OR |
| 9b. | Unit – I |
| 10a. | Unit – II |
| | OR |
| 10b. | Unit – II |
| 11a. | Unit – III |
| | OR |
| 11b. | Unit – III |
| 12a. | Unit – IV |
| | OR |
| 12b. | Unit – IV |